event_name	Sum of magnitude	Sum of max_mmi	Sum of population_mmi_ge_6
christchurch_feb_2011	6.20	9.40	461,764.88
christchurch_jun_2011	6.00	8.80	406,498.80
cook_strait_2013	6.50	7.60	19,949.94
darfield_2010	7.20	8.40	534,594.03
east_cape_2016	7.10	5.20	0.00
eketahuna_2014	6.20	6.80	84,416.14
kaikoura_2016	7.80	8.40	83,276.46
kermadec_2021	7.90	2.20	0.00
lake_grassmere_2013	6.50	8.00	31,464.04
taumarunui_deep_2026	5.50	2.80	0.00
Total	**66.90**	**67.60**	**1,621,964.29**

event_name	origin_time	Sum of magnitude	Sum of max_mmi	event_time_status
christchurch_feb_2011	2011/2/21 23:51:42	6.20	9.40	not_supported
christchurch_jun_2011	2011/6/13 2:20:49	6.00	8.80	not_supported
cook_strait_2013	2013/7/21 5:09:30	6.50	7.60	not_supported
darfield_2010	2010/9/3 16:35:41	7.20	8.40	not_supported
east_cape_2016	2016/9/1 16:37:55	7.10	5.20	not_supported
eketahuna_2014	2014/1/20 2:52:45	6.20	6.80	not_supported
kaikoura_2016	2016/11/13 11:02:56	7.80	8.40	not_supported
kermadec_2021	2021/3/4 19:28:32	7.90	2.20	supported
lake_grassmere_2013	2013/8/16 2:31:05	6.50	8.00	not_supported
taumarunui_deep_2026	2026/10/15 5:25:34	5.50	2.80	supported
Total		**66.90**	**67.60**	

event_name	Sum of population_mmi_ge_6	Sum of as_if_today_buildings
christchurch_feb_2011	461,764.88	3219311
christchurch_jun_2011	406,498.80	3219311
cook_strait_2013	19,949.94	3219311
darfield_2010	534,594.03	3219311
east_cape_2016	0.00	3219311
eketahuna_2014	84,416.14	3219311
kaikoura_2016	83,276.46	3219311
kermadec_2021	0.00	3219311
lake_grassmere_2013	31,464.04	3219311
taumarunui_deep_2026	0.00	3219311
Total	**1,621,964.29**	**32193110**

event_name	Sum of as_if_today_buildings	Sum of buildings_mmi_ge_6
christchurch_feb_2011	3219311	270518
christchurch_jun_2011	3219311	236306
cook_strait_2013	3219311	16059
darfield_2010	3219311	337971
east_cape_2016	3219311	0
eketahuna_2014	3219311	65461
kaikoura_2016	3219311	78564
kermadec_2021	3219311	0
lake_grassmere_2013	3219311	26492
taumarunui_deep_2026	3219311	0
Total	**32193110**	**1031371**

event_name	event_time_status	Sum of validation_mae
christchurch_feb_2011	not_supported	0.86
christchurch_jun_2011	not_supported	0.85
cook_strait_2013	not_supported	0.85
darfield_2010	not_supported	1.16
east_cape_2016	not_supported	1.14
eketahuna_2014	not_supported	0.64
kaikoura_2016	not_supported	1.37
kermadec_2021	supported	2.82
lake_grassmere_2013	not_supported	0.79
taumarunui_deep_2026	supported	2.31
Total		**12.79**

Limitations and quality notes
1. LIMITATIONS
2. No policy data: exposure uses building count, footprint area, and population as proxies.
3. Shaking intensity is indicative only; it is not actuarial or engineering advice.
4. Building appearance and disappearance dates are approximate and inferred from imagery.